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The Mean Value Inequality

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Definition 1. PolygonalLine Let $(E, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$, with subset $A \subseteq E$. A *polygonal line* in A is a sequence of $n + 2$ points $u_0, \dots, u_{n+1}$ in E , such that each line segment connecting neighbouring vertices lies completely in A , i.e.

$$t \cdot u_i + (1 - t) \cdot u_{i+1} \in A \quad \forall i = 0, \dots, n \text{ and } t \in [0, 1].$$

Lemma 2. PolygonalLine.trans def:polygonalLine Let U be a subset of a normed vector space. Given a polygonal line in U , joining $x \in U$ and $y \in U$, together with another polygonal line joining y and $z \in U$, then by connecting the two polygonal lines head-to-tail, the result is a polygonal line in U , joining x and z .

Proof. The construction of the composition of polygonal lines follows common sense: For the polygonal lines a , defined by $(x, a_1, \dots, a_n, y)$, and b , defined by $(y, b_1, \dots, b_m, z)$, we define the composition $a \circ b$ by the sequence $(x, a_1, \dots, a_n, y, b_1, \dots, b_m, z)$.

It remains to show that each segment is still contained in U . This is trivial as the first $n + 2$ vertices belong to a and are all contained in U . Similarly, the last $m + 2$ vertices belong to b , and are all contained in U as well. $\square$

Theorem 3. polygonal_connected_of_connectedlem : composition_of_polygonal_line_in_U_is_an_open_connected_set_of_a

Proof. The case where $U = \emptyset$ is trivial. So we assume U to be non-empty. Fix a point $x \in U$, and consider the set

$$V := \{u \in U \mid \exists \text{ a polygonal line connecting } x \text{ and } u.\}$$

We want to show that $U = V$.

To show that, we prove that V is both closed and open in U and V is non-empty. The latter is true, as x must be in V .

For closedness, we show that $V = \bar{V}$, the closure of V . Fix a point $a \in \bar{V}$, then $\exists \epsilon > 0$, such that $B_\epsilon(a) \cap V \neq \emptyset$. Take a point $b \in B_\epsilon(a) \cap V$. Since $b \in V$, there is a polygonal line joining x and b . Since $b \in B_\epsilon(a)$, there is a line segment joining b and a , which is a polygonal line as well. By composing the two polygonal lines, we create a polygonal line joining x and a . Hence $a \in V$.

For openness, fix a point $a \in V$. We take an open ball centred at a and contained in U , as U is open. It can be shown that this open ball is contained in V as well. It suffices to show that every point b in the open ball can be connected to x using a polygonal line, which can be constructed by composing the polygonal line joining x and a with the line segment joining a and b .

Since V is both open and closed in U , it is either equal to U or is the empty set. Since it is non-empty, we must have $U = V$. $\square$

Definition 4. thm:existence_of_polygonal_line Let U be an open connected set of a Banach space E . For $a, b \in U$, we define $d_U(a, b)$ to be the infimum of the lengths of the polygonal lines contained in U joining a and b .

Theorem 5. *def:distance in U lem:composition_of_polygonal_line_thm : polygonal_line_length_at_least_interval_length is a distance in the topological space U.*

Proof. Let φ be an arbitrary polygonal line joining a to b in U , defined by the sequence $a_0, \dots, a_{n+1}$, with $a_0 = a$ and $a_{n+1} = b$

Positivity: The length of φ is $\text{Length}(\varphi) := \sum_{i=0}^n \|a_{i+1} - a_i\|$, which by the definition of a norm function is strictly non-negative. Since $d_U(a, b)$ is the infimum of a strictly non-negative set, it must itself be non-negative. If $d_U(a, b) = 0$, by the previous theorem we have $\|b - a\| \leq d_U(a, b) = 0$, which implies $a = b$.

Symmetry: For any polygonal line φ from a to b in U , the reverse line ψ defined by $a_{n+1}, \dots, a_0$ is a polygonal line from b to a . $\text{Length}(\psi) = \sum_{i=0}^n \|a_i - a_{i+1}\| = \sum_{i=0}^n \|a_{i+1} - a_i\| = \text{Length}(\varphi)$, so the set of lengths of polygonal lines from a to b and from b to a are equivalent and therefore $d_U(a, b) = d_U(b, a)$.

Triangle Inequality: Let φ be a polygonal line from a to b and ψ be a polygonal line from b to c . The composition of φ and ψ is a polygonal line θ from a to c , where

$$\begin{cases} \varphi \text{ is defined by } (a, a_1, \dots, a_n, b) \\ \psi \text{ is defined by } (b, b_1, \dots, b_m, c) \\ \theta \text{ is defined by } (a, a_1, \dots, a_n, b, b_1, \dots, b_m, c) \end{cases}$$

By the definition of infimum, $d_U(a, c) = \inf \text{Length}(\theta) \leq \text{Length}(\theta) = \text{Length}(\varphi) + \text{Length}(\psi)$. Since φ and ψ are arbitrary, we take the infimum on the right hand side, and hence $d_U(a, c) \leq d_U(a, b) + d_U(b, c)$. $\square$

Theorem 6. *def:distance in U thm:existence_of_polygonal_line_norm_image_sub_epsilon_norm_has_F_Deriv_Within_epsilon' Let $U \rightarrow F$ be a differentiable mapping with values in a Banach space F . Assume that*

$$\|f'(x)\| \leq k \quad \forall x \in U$$

then for any $x_1, x_2 \in U$, $\|f(x_1) - f(x_2)\| \leq k \cdot d_U(x_1, x_2)$.

Proof. Let $x_1, x_2 \in U$ and l a polygonal line contained in U joining x_1 and x_2 . Suppose l is defined by $n + 1$ vertices $a_0, \dots, a_{n+1}$, where $a_0 = x_1, a_{n+1} = x_2$ and $a_i \in U$. Then from the mean value inequality on each interval (a_i, a_{i+1}) , we have $\|f(a_i) - f(a_{i+1})\| \leq k \cdot \|a_i - a_{i+1}\|$. From the triangle inequality we have

$$\begin{aligned} \|f(x_1) - f(x_2)\| &\leq \sum_{i=0}^n \|f(a_i) - f(a_{i+1})\| \\ &\leq \sum_{i=0}^n k \cdot \|a_i - a_{i+1}\| \\ &= k \cdot \text{Length}(l) \end{aligned}$$

as $\text{Length}(l)$ is the sum of the length of each interval. Therefore, taking the infimum we get

$$\|f(x_1) - f(x_2)\| \leq k \cdot \inf \text{Length}(l) = k \cdot d_U(x_1, x_2)$$

□

Definition 7. Let $(E, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$. The *length of an interval* with origin $a \in E$ and end $b \in E$ is $d(a, b) := \|b - a\|$.

Definition 8. `def:polygonalLine` `def:intervalLength` Given a polygonal line φ defined by $(a_0, \dots, a_{n+1})$, with some fixed $n \geq 0$ the *length of the polygonal line* φ is the sum of the lengths of the intervals from which it is formed:

$$\text{Length}(\varphi) := \sum_{i=0}^n d(a_i, a_{i+1}) = \sum_{i=0}^n \|a_{i+1} - a_i\|$$

Lemma 9. Given a sequence $(a_i)_{i=0}^n$ with $a_i \in E$ for some $n \geq 1$, $\sum_{i=0}^{n-1} \|a_{i+1} - a_i\| \geq \|a_n - a_0\|$

Proof. • **Base Case** ($n = 1$): For $n = 1$, we need $\|a_1 - a_0\| \geq \|a_1 - a_0\|$ which follows immediately by reflexivity.

- **Induction Hypothesis:** Assume that the result holds for any such sequence $(b_i)_{i=0}^k$. That is,

$$\sum_{i=0}^{k-1} \|b_{i+1} - b_i\| \geq \|b_k - b_0\|$$

- **Induction Step:** We want to show that $\sum_{i=0}^k \|a_{i+1} - a_i\| \geq \|a_{k+1} - a_0\|$ follows from this assumption. For some arbitrary sequence $(a_i)_{i=0}^{k+1}$,

$$\begin{aligned} \|a_{k+1} - a_0\| &= \|(a_{k+1} - a_k) + (a_k - a_0)\| \\ &\leq \|a_{k+1} - a_k\| + \|a_k - a_0\| \\ &\leq \|a_{k+1} - a_k\| + \sum_{i=0}^{k-1} \|a_{i+1} - a_i\| \\ &= \sum_{i=0}^k \|a_{i+1} - a_i\| \end{aligned}$$

Therefore, by the principle of mathematical induction, $\sum_{i=0}^{n-1} \|a_{i+1} - a_i\| \geq \|a_n - a_0\|$ for any such sequence with any $n \geq 1$. □

Theorem 10. *lemma: norm_sum_atleast_sum_norm* The length of a polygonal line φ defined by $a_0, \dots, a_{n+1}$ for some fixed $n \geq 0$ between origin $a = a_0$ and end $b = a_{n+1}$ is at least equal to the length of the interval between them:

$$\sum_{i=0}^n \|a_{i+1} - a_i\| \geq \|b - a\|$$

Proof. The result follows immediately from the lemma.

□